

UBIFEED-500 POWDER

Ingredient(s):

Each g contains :	
Chlortetracycline (as Chlortetracycline HCl)	88mg
Penicillin (as Penicillin G Procaine)	44mg
Sulfathiazole	88mg

Pharmacology (Summary of Pharmacodynamics and Pharmacokinetics):

Pharmacodynamics:

Chlortetracycline -

Tetracyclines generally act as bacteriostatic antibiotics and inhibit protein synthesis by reversibly binding to 30s ribosomal subunits of susceptible organisms, thereby preventing binding to those ribosomes of aminoacyl transfer-RNA. Tetracyclines also are believed to reversibly bind to 50s ribosomes and additionally alter cytoplasmic membrane permeability in susceptible organisms. In high concentrations, tetracyclines can also inhibit protein synthesis by mammalian cells..

Penicillin G Procaine

Penicillin G procaine also known as procaine penicillin G or procaine benzylpenicillin. Penicillins are usually bactericidal against susceptible bacteria and act by inhibiting mucopeptide synthesis in the cell wall resulting in a defective barrier and an osmotically unstable spheroplast. The exact mechanism for this effect has not been definitely determined, but beta-lactam antibiotics have been shown to bind to several enzymes (carboxypeptidases, transpeptidases, endopeptidases) within the bacterial cytoplasmic membrane that are involved with cell wall synthesis. The different affinities that various beta-lactam antibiotics have for these enzymes (also known as penicillin-binding proteins; PBPs) help explain the differences in spectrums of activity the drugs have that are not explained by the influence of beta-lactamases. Like other beta-lactam antibiotics, penicillins are generally considered more effective against actively growing bacteria.

Sulfathiazole –

The bacteriostatic action of sulfonamides is through effects on essential bacterial metabolites. The structural similarity between sulfonamide and p-aminobenzoic acid (PABA) and postulated that the two compounds competed for a critical enzymatic site. Inhibition of bacterial growth resulted from reduced activity which effected by the sulfonamide-bound enzyme. The enzyme-catalyzed condensation of PABA and 2-amino-4-hydroxytetrahydropteridine, a necessary step in synthesis of folic acid, was inhibited by sulfonamides. Sulfonamide-sensitive bacteria are unable to utilize a preformed source of folic acid. Animal cells have an absolute requirement for preformed folic acid. Lack of folic acid synthesis in the animal host may explain the therapeutic success obtained with sulfonamides, since the drugs selectively inhibit bacterial synthesis of folic acid without producing concomitant effects in host tissue.

Pharmacokinetics:

Chlortetracycline -

Following oral administration, tetracyclines are absorbed readily from the stomach and the small intestine to give a peak plasma concentration in 2–4 hours in carnivorous animals. Calcium and magnesium ions in the GI tract will diminish absorption. Some of the drug is concentrated by the liver, excreted in bile, and resorbed from the intestines, so a small amount persists in the blood as a result of enterohepatic resorption. Recommended dosages of tetracycline produce plasma concentrations during most of the 24 hours postadministration. Tetracyclines undergo metabolism to various degrees. The most frequently identified substance in urine, faeces, and tissue is the parent tetracycline. As much as 30% will be excreted unchanged in faeces. Tetracyclines are reversibly bound to plasma proteins and are widely distributed. They are removed from blood by the liver, and high concentrations are achieved in parenchyma and bile. Bile concentration may be 30 times that of blood; however, enterohepatic recirculation limits bile secretion and prolongs maintenance of therapeutic concentrations.

Tetracyclines diffuse throughout the body and are found in highest concentrations in kidney, liver, spleen, and lung. They are also deposited at active sites of ossifications. Sites of growth in the diaphysis and epiphysis can be identified by the fluorescence produced by tetracyclines. Concentration in fetal blood is approximately one-half that in maternal blood. Tetracyclines diffuse into the cerebrospinal fluid with difficulty. Tetracyclines also pass through the bovine placenta and enter fetal circulation, as well as prostatic fluid. Tetracyclines are incorporated in chicken and turkey eggshells. The presence of calcium and other ions in the milk substitute decreases absorption because of the chelating action of chlortetracycline.

Tetracyclines are excreted primarily by the kidneys. Approximately 25–30% of a single dose can be found in urine. Following oral administration, the highest urine concentrations is between 2 and 8 hours, but antibacterial activity can be detected for 3 days or more after therapy is discontinued. Faecal elimination of tetracyclines occurs regardless of route of administration. The amount eliminated following oral administration may reach 10% of the dose. Tetracycline antibiotics are also eliminated in milk; concentration is about one-half that of maternal serum.

Penicillin G Procaine –

The oral absorption characteristics of the penicillins are dependent upon its class. Penicillin G is the only available oral penicillin that is substantially affected by gastric pH and can be completely inactivated at pH's of less than 2. Penicillins are generally distributed widely throughout the body. Most drugs attain therapeutic levels in the kidneys, liver, heart, skin, lungs, intestines, bile, bone, prostate, and peritoneal, pleural and synovial fluids. Penetration into CSF and eye only occur with inflammation and may not reach therapeutic levels. Penicillins are bound in varying degrees to plasma proteins and they cross the placenta. Most penicillins are rapidly excreted largely unchanged by the kidneys into the urine via glomerular filtration and tubular secretion. Probenedic can

prolong half-lives and increase serum levels by blocking the tubular secretion of penicillins. Except for nafcillin and oxacillin, hepatic inactivation and biliary secretion is a minor route of excretion.

Sulfathiazole –

Absorption of a sulfonamide and passage from its site of administration to blood are governed by passive diffusion. The absorption rate of the solubilized drug is influenced by several factors, including ionization state and lipophilicity of the drug, vascularity of the absorption site, and extent to which the drug is bound to material present at the administration site. Although most sulfonamides are rapidly absorbed from the GI tract, some interspecies variation occurs. Sulfathiazole is slowly absorbed by sheep and cattle. In general, cattle absorb sulfonamides from the GI tract more slowly than sheep or swine. Because of the slow absorption and rapid excretion of sulfathiazole by cattle and sheep, it is very difficult to establish and maintain blood concentrations of the drug exceeding 2mg/dL in these species. Effects of age, dosage form, and environment on absorption of sulfonamides following oral administration have been characterized. When sulfathiazole is administered in a bolus to calves ranging in age from 1–16 weeks, the younger animals absorb the drug more rapidly. Absorption of sulfonamides by diseased animals may be quite different from that observed in healthy individuals of the same species.

Once sulfonamides have been absorbed, they undergo rapid mixing in the bloodstream. Subsequent movement of these drugs into other biologic fluids and tissues is influenced by the ionization state of the sulfonamide, vascularity of specific tissues, presence of specific barriers to sulfonamide diffusion, and fraction of the administered dose bound to plasma proteins. Distribution of sulfonamides into various tissues and fluids occurs as a result of passive diffusion. In the absence of active transport processes or ion-trapping phenomena, concentration of sulfonamides in various fluids and tissues is a reflection of the rate at which the drug equilibrated between blood and other body components. Sulfonamide in blood is bound to plasma proteins, primarily albumin. The unbound drug fraction is freely diffusible. The ionization state of a sulfonamide has a profound influence upon further distribution once the drug has crossed the capillary endothelium. Unbound, nonionized drugs cross biologic membranes more rapidly than their bound or ionized forms. A dynamic equilibrium is established between the sulfonamide in blood, other extracellular fluids, and intracellular fluids. Plasma-tissue equilibrium is established more rapidly in highly vascularized tissues. Concentrations of sulfonamides in tissues are consistently lower than those observed in plasma. The blood-brain barrier is permeable to the non-ionized fraction of freely diffusible sulfonamides. As a result, concentration of these compounds in cerebrospinal fluid is equivalent to concentration of unbound and nonionized drugs in blood.

Sulfonamides are extensively metabolized in the animal body. Sulfonamide is acetylated depends upon the drug administered as well as the recipient species. In general, acetylated sulfonamides are the major urinary excretion product following enteral or parenteral administration of sulfonamides to cattle, sheep, and swine. Oxidation of the benzene and heterocyclic rings of sulfonamides is an important metabolic pathway in certain species (sheep and cattle). Sulfonamides and their metabolites are often conjugated with glucuronic acid and sulfates. Conjugation often occurs at sites of previous hydroxylation (benzene ring, heterocyclic ring, or aliphatic constituents of heterocyclic rings) or at the N1 or N4 positions. The amount of drug eliminated in urine as glucuronides or sulfates is dependent upon the sulfonamide administered and the species to which it is given. Ring cleavage appears to be a metabolic pathway of minor importance in most animals, as only small amounts of cleavage products are excreted in urine. In general, acetylated, hydroxylated, and conjugated forms of the sulfonamides exhibit marked decrease in antibacterial activity when compared to their parent compounds.

Sulfonamides are eliminated in urine, faeces, bile, milk, sweat, and tears. The kidney is primarily involved in excretion of these drugs. The rate of sulfonamide excretion is primarily determined by the net consequences of glomerular filtration, renal tubular resorption, and active tubular secretion. All sulfonamides existing in an ultrafilterable state in plasma are filtered from the blood at the renal glomerulus. Rate of filtration is proportional to concentration of unbound drug and metabolites in blood. The amount of a sulfonamide available for resorption (nonionized form) by passive diffusion is dependent on the pH of fluids within the tubular lumen. Increased persistence of the drug in the body is attributed to renal processes of resorption that recycle filtered drug back into blood. Increased urine flow decreases the rate of sulfonamide resorption. At higher urine flow rates, concentration of diffusible drug is reduced (dilution effect), which subsequently reduces the rate of passive drug diffusion across the tubular epithelium. Certain sulfonamides are secreted into renal tubular fluid by an active transport process. Substrate specificity is dependent of the sulfonamide. Tubular secretion of sulfathiazole occurs in the horse; other reports indicate that the drug evidences similar behavior in swine and cattle.

Indication:

Treatment of bacterial enteritis (Salmonellosis or necrotic enteritis caused by *Salmonella choleraesuis* and vibronic dysentery).

Target Species

Swine.

Dosage and Administration:

Mix 1.2kg of Ubifeed-500 Powder into every ton (907.2kg) of feed.
Orally via mixing with feed.

Contraindication(s):

- 1) Chlortetracycline is contraindicated in patients hypersensitive to it or other tetracyclines.
- 2) Penicillins are contraindicated in patients who have a history of hypersensitivity to them. Because there may be cross-reactivity, use penicillin cautiously in patients who are documented hypersensitive to other beta-lactam antibiotics (e.g., cephalosporins, cefamycins, carbapenems).
- 3) Fluid therapy for treatment of renal damage produced by sulfonamides is contraindicated in anuric. animals because of the attendant risk of overhydration.
- 4) Other adverse reactions to sulfonamides in dogs include keratoconjunctivitis sicca, bone marrow suppression, and cutaneous allergic reactions.
- 5) Disorders of the hematopoietic system have been observed following use of sulfonamides for treatment of animal disease.

Warning and Precautions

- 1) In patients with renal insufficiency or hepatic impairment, chlortetracycline must be used cautiously. Lower than normal dosages are recommended with enhanced monitoring of renal and hepatic function. Avoid concurrent administration of other nephrotoxic or hepatotoxic drugs if tetracyclines are administered to these patients.
- 2) Do not administer systemic antibiotics orally in patients with septicemia, shock, or other grave illness as absorption of the medication from the GI tract may be significantly delayed or diminished.
- 3) Certain species (snakes, birds, turtles, guinea pigs, and chinchillas) are reportedly sensitive to procaine penicillin G.

Interaction with Other Medicament(s):

- 1) When orally administered, tetracyclines can chelate divalent or trivalent cations that can decrease the absorption of the tetracycline or the other drug if it contains these cations. Oral antacids, saline cathartics or other GI products containing aluminium, calcium, magnesium, zinc or bismuth cations are most commonly associated with this interaction.
- 2) Oral iron products are also associated with decreased tetracycline absorption.
- 3) Oral sodium bicarbonate, kaolin, pectin, or bismuth subsalicylate may impair tetracycline absorption when given together orally.
- 4) Bacteriostatic drugs, like the tetracyclines, may interfere with bactericidal activity of the penicillins, cephalosporins, and aminoglycosides. There is some amount of controversy regarding the actual clinical significance of this interaction, however.
- 5) Tetracyclines may increase the bioavailability of digoxin in a small percentage of patients (human) and lead to digoxin toxicity.
- 6) Tetracyclines may depress plasma prothrombin activity and patients on anticoagulant (e.g., warfarin) therapy may need dosage adjustment.
- 7) Tetracyclines have been reported to increase the nephrotoxic effects of methoxyflurane and tetracycline HCl or oxytetracycline are not recommended for use with methoxyflurane.
- 8) GI side effects may be increased if tetracyclines are administered concurrently with theophylline products.
- 9) Tetracyclines have reportedly reduced insulin requirements in diabetic patients, but this interaction is yet to be confirmed with controlled studies.
- 10) In vitro studies have demonstrated that penicillins can have synergistic or additive activity against certain bacteria when used with aminoglycosides or cephalosporins.
- 11) Use of bacteriostatic antibiotics (e.g., chloramphenicol, erythromycin, tetracyclines) with penicillins is generally not recommended, particularly in acute infections where the organism is proliferating rapidly as penicillins tend to perform better on actively growing bacteria.
- 12) In low concentrations, certain penicillins (e.g., ampicillin, oxacillin or nafcillin) may have additive or synergistic effects against certain bacteria when used with rifampin, but there is apparent antagonism when the penicillin is present in high concentrations.
- 13) Probenecid competitively blocks the tubular secretion of most penicillins, thereby increasing serum levels and serum half-lives.
- 14) High dosages of certain penicillins (e.g., ticarcillin, carbenicillin) have been associated with bleeding; they should be used cautiously in patients receiving oral anticoagulants or heparin.

Pregnancy and Lactation

- 1) Because tetracyclines can retard fetal skeletal development and discolor deciduous teeth, they should only be used in the last half of pregnancy when the benefits outweigh the fetal risks.
- 2) Penicillins have been shown to cross the placenta and safe use of them during pregnancy has not been firmly established, but neither have there been any documented teratogenic problems associated with these drugs. However, use only when the potential benefits outweigh the risks.
- 3) Penicillins are excreted in maternal milk in low concentrations; use may cause diarrhea, candidiasis or allergic response in nursing offspring.

Side Effect(s)/Adverse Reaction(s):

1. Chlortetracycline given to young animals can cause discoloration of bones and teeth to a yellow, brown, or gray color. High dosages or chronic administration may delay bone growth and healing.
2. Tetracyclines in high levels can exert an antianabolic effect that can cause an increase in BUN and/or hepatotoxicity, particularly in patients with preexisting renal dysfunction. As renal function deteriorates secondary to drug accumulation, this effect may be exacerbated.
3. In ruminants, high oral doses can cause ruminal microflora depression and ruminoreticular stasis.
4. In small animals, tetracyclines can cause nausea, vomiting, anorexia and diarrhea. Cats do not tolerate oral tetracycline or oxytetracycline very well, and may also present with symptoms of colic, fever, hair loss and depression. There are reports that long-term tetracycline use may cause urolith formation in dogs.
5. Horses that are stressed by surgery, anesthesia, trauma, etc., may break with severe diarrheas after receiving tetracyclines (especially with oral administration).
6. Tetracycline therapy (especially long-term) may result in overgrowth (superinfections) of non-susceptible bacteria or fungi.
7. Tetracyclines have also been associated with photosensitivity reactions and, rarely, hepatotoxicity or blood dyscrasias.
8. Hypersensitivity reactions unrelated to dose can occur with these penicillins and can be manifested as rashes, fever, eosinophilia, neutropenia, agranulocytosis, thrombocytopenia, leukopenia, anemias, lymphadenopathy, or full-blown anaphylaxis.
9. When given orally, penicillins may cause GI effects (anorexia, vomiting, diarrhea). Because the penicillins may also alter gut flora, antibiotic-associated diarrhea can occur, as well as selecting out resistant bacteria maintaining residence in the colon of the animal (superinfections).
10. High doses or very prolonged use has been associated with neurotoxicity (e.g., ataxia in dogs). Although the penicillins are not considered hepatotoxic, elevated liver enzymes have been reported. Other effects reported in dogs include tachypnea, dyspnea, edema and tachycardia.
11. Disturbances of the kidney and urinary tract may occur following administration of sulfonamides. Crystalluria, hematuria, and obstruction of renal tubules are the commonest sequelae. Decreased water consumption and aciduria increase the risk that crystallization will occur. Anorexia, depression, hematuria, crystalluria, renal colic, frequent attempts to urinate, and elevation of blood urea nitrogen may be observed following renal presipitation of sulfonamides.

Environmental Property

None has been mentioned in animal.

Symptoms and Treatment of Overdosage, and Antidote(s):

Oral overdoses of chlortetracycline would most likely be associated with GI disturbances (vomiting, anorexia, and/or diarrhea). Should the patient develop severe emesis or diarrhea, fluids and electrolytes should be monitored and replaced if necessary.

Storage Condition(s):

Store below 30°C.

Product Description and Packing(s):

A pale yellowish-brown color powder, and without other foreign matter.
Aluminum Pouch of 100g, 1kg. Paper Bag of 20Kg

Shelf Life:

4 years

For food producing animals product:

Withdrawal Periods

7 days before slaughter.

Maximum Residual Limit (MRL)

Chlortetracycline: Pig: Muscle 0.2mg/kg, liver 0.6mg/kg, kidney 1.2mg/kg

Procaine penicillin G: 0 µg/kg

Sulfathiazole: Muscle, fat, liver, kidney 100µg/kg



Made by YSP

Manufacturer:
Y.S.P INDUSTRIES VIETNAM CO., LTD
Road No.6, Nhon Trach Industrial Zone 3, Nhon Trach Dist ,
Dong Nai Province, Vietnam Tel: +84251 3569 993

Product Registration Holder :
Y.S.P. INDUSTRIES (M) SDN. BHD.
Lot 3, 5 & 7, Jalan P/7, Section 13,
Kawasan Perindustrian Bandar Baru Bangi,
43000 Kajang, Selangor Darul Ehsan Malaysia.
Product Info: 1 800 88 3679

Date of revision: 18Jan21

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